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Data replication and migration

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Data replication and migration

01	Replication and migration architecture
02	The `gcloud` command line tool
03	Moving datasets
04	Datastream



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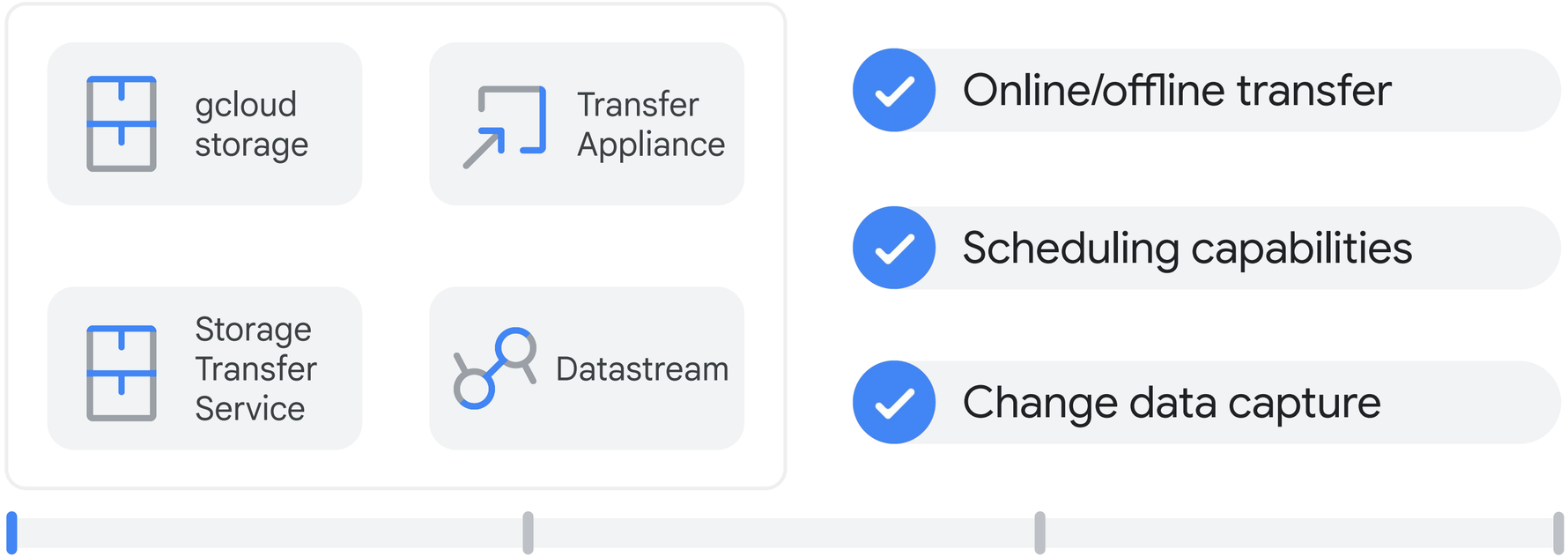
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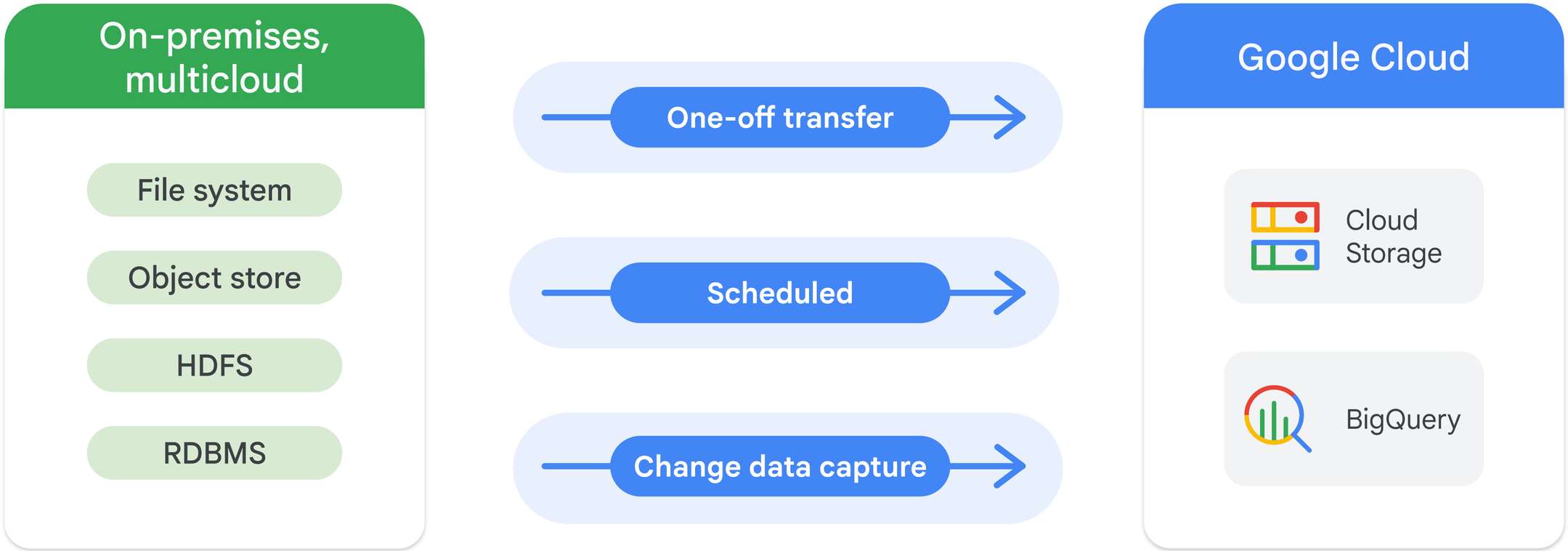
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Replication and migration services onboard your data into Google Cloud



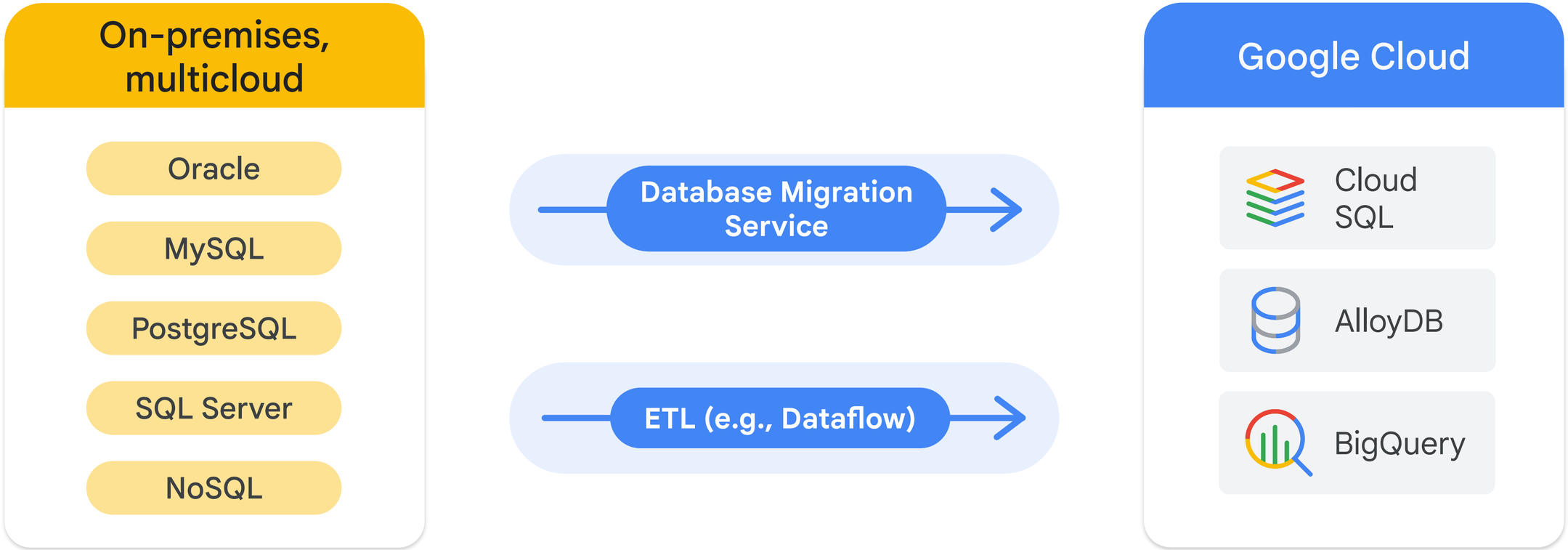
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Common replication and migration scenarios for ingesting your data into Google Cloud



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Google Cloud provides additional options for migrating your workloads



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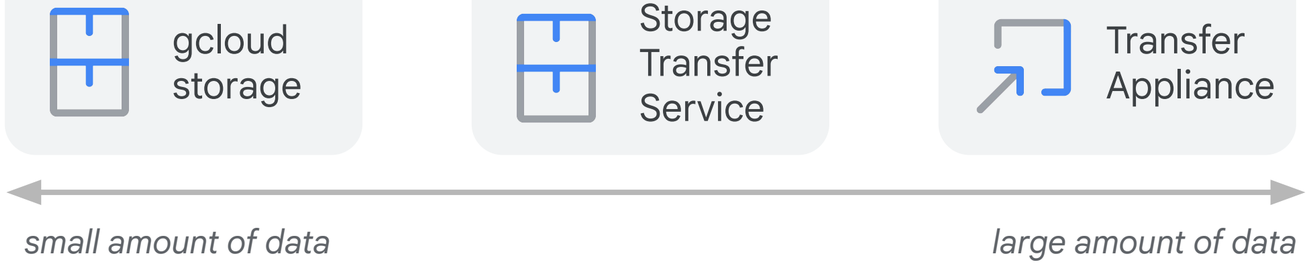
Choosing between migration options depends on your amount of data and network bandwidth

1 TB + 100 Gbps

2 minutes

1 TB + 100 Mbps

30 hours



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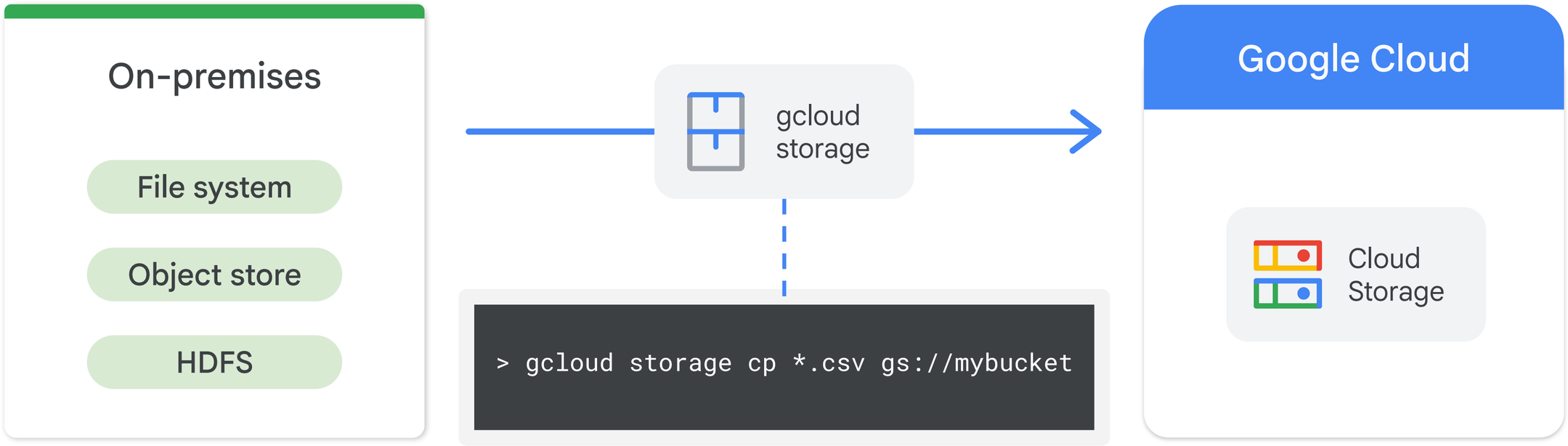
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Use the `gcloud storage` command to move small amounts of data to Cloud Storage



- Small to medium-sized transfers
- Executed on an as-needed basis

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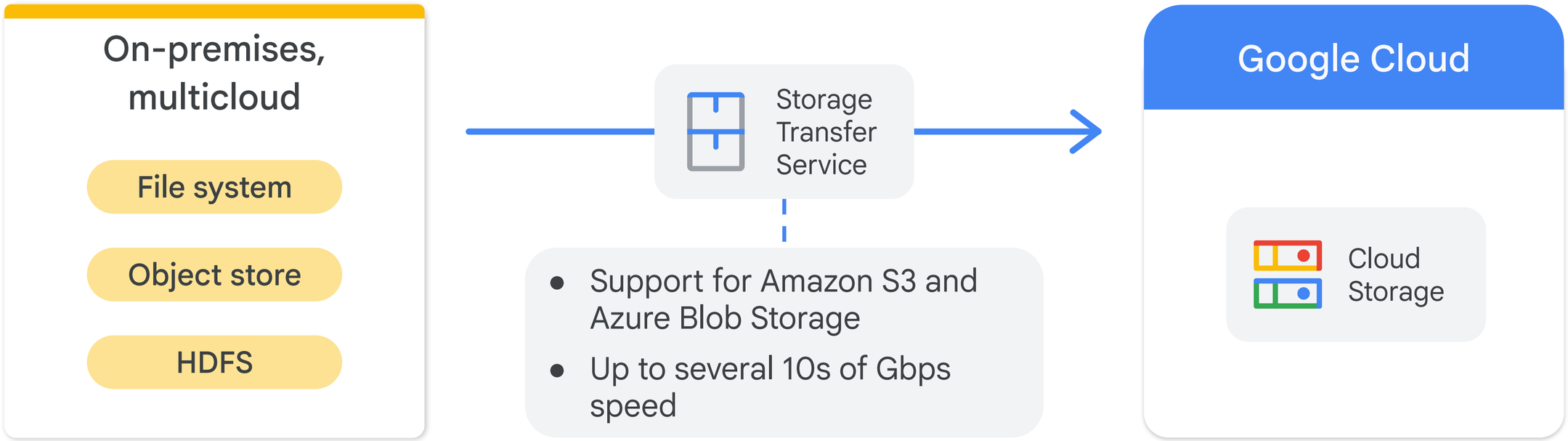
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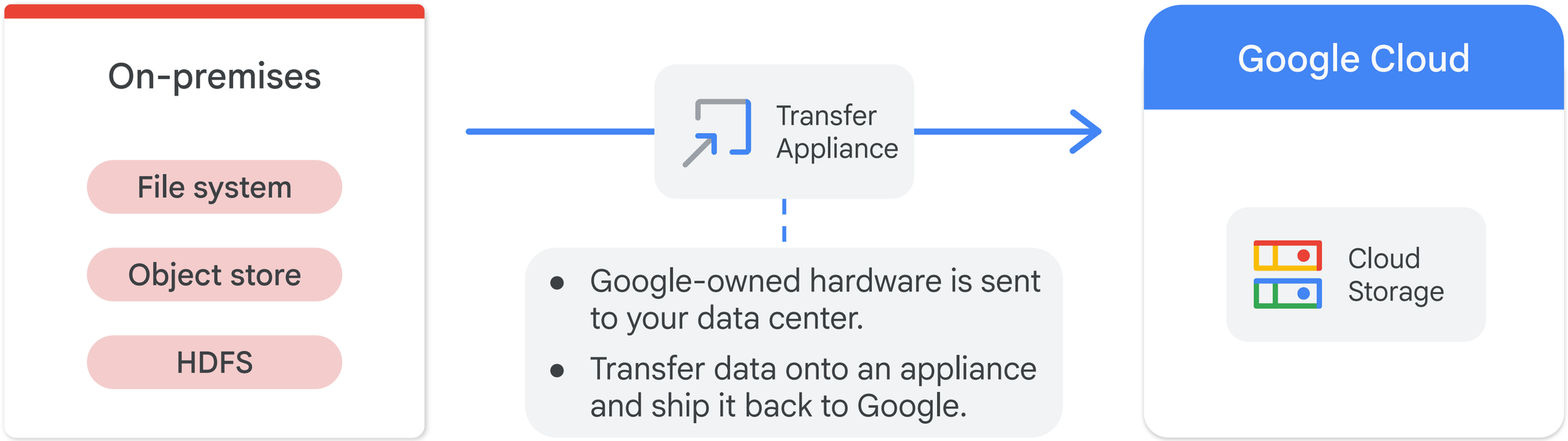
Move large amounts of data with Storage Transfer Service



- Medium to large-sized transfers
- Scheduling capabilities

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Use Transfer Appliance to move large amounts of data in offline mode



- Large transfers with limited bandwidth
- Multiple appliance sizes available

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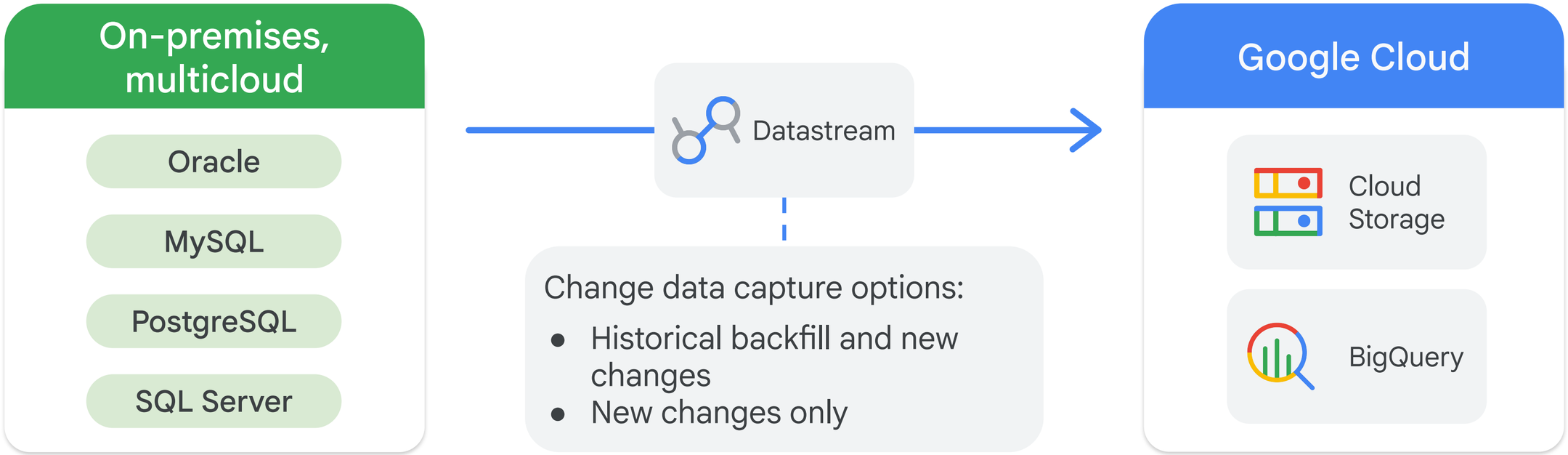
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Datastream continuously replicates your RDBMS into Google Cloud for analytics



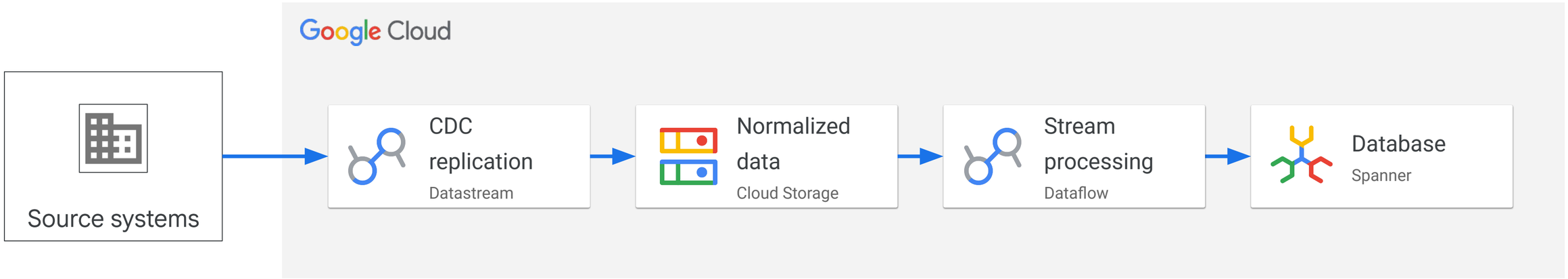
Public or private connectivity options

Include or exclude schema, table, column

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Datastream use cases

- Analytics with database replication into BigQuery
- Analytics with custom data processing in Dataflow into BigQuery
- Data processing using an event-driven architecture
- **Example:** database replication and migration using Dataflow templates



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Datastream uses the database's write-ahead log (WAL) to process change events

Event in RDBMS

- INSERT
- UPDATE
- DELETE



Event in WAL

- LogMiner (Oracle)
- Binary log (MySQL)
- Logical decoding (PostgreSQL)
- Transaction logs (SQL Server)

Event processing

Datastream

Event

Payload

Event is stored

- AVRO
- JSON
- Table

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Compare migration and replication options

	gcloud storage	Storage Transfer Service	Transfer Appliance	Datastream
Transfer type	online	online	offline	online
Source locations	On-premises, Google Cloud	On-premises, Google Cloud, multicloud	On-premises	On-premises, Google Cloud, multicloud
Data range	less than 1 TB recommended	more than 1 TB recommended	7 TB, 40 TB, 300 TB	10,000 tables per stream
Velocity	Batch	Batch (hourly at minimum)	Batch	Batch and Stream
Data format	any	any	any	structured

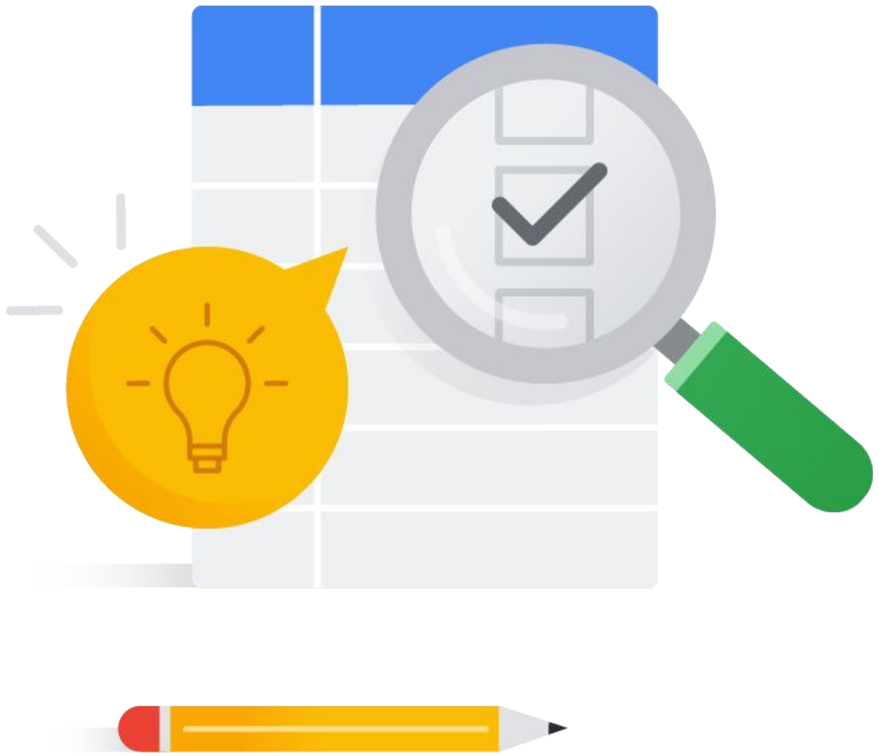
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Lab (optional): Datastream: PostgreSQL Replication to BigQuery

45 min

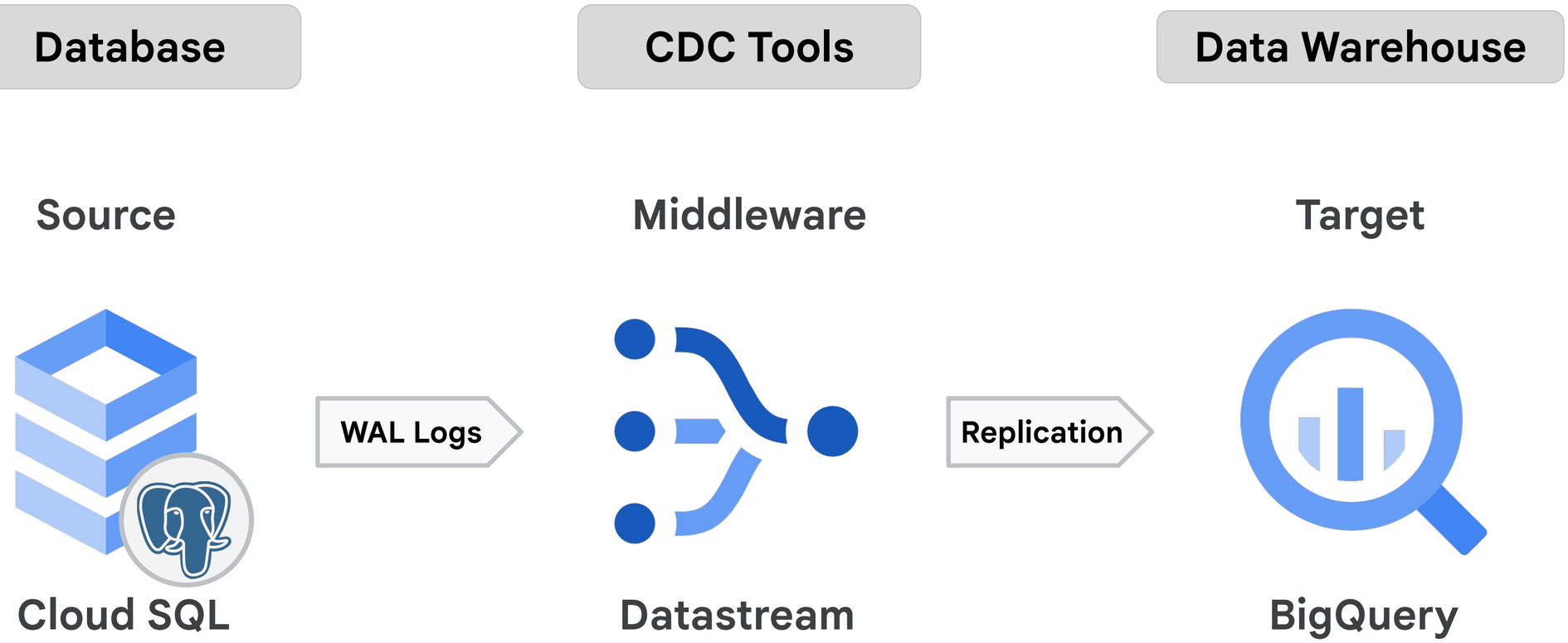
Learning objectives

- Prepare a Cloud SQL for PostgreSQL instance using the Google Cloud console.
- Import data into the Cloud SQL instance.
- Create a Datastream connection profile for the PostgreSQL database.
- Create a Datastream connection profile for the BigQuery destination.
- Create a Datastream stream and start replication.
- Validate that the existing data and changes are replicated correctly into BigQuery.



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Lab (optional): Datastream: PostgreSQL Replication to BigQuery




Google Cloud